

Quiz 1 – Study Guide

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Module 01: Law and Accessibility

Explain the roles of ADA Title II, Section 504, and Section 508, and how legal requirements relate to technical standards.

ADA Title II: Requires STATE and LOCAL GOVERNMENTS to ensure that ALL public services are fully accessible to individuals with disabilities.

Section 504: Section 504 is a broad scope that ensures that all organizations under federal funding must protect anyone who has a disability from an unfair experience.

Section 508: Federal law that requires the U.S. government agencies to ensure their information and communication technology is fully accessible to people with disabilities.

Relation: The law sets the stage for an obligation in the U.S. They state who must provide access and what general civil rights protections apply. The law gets intertwined with technical standards to ensure all users have an equal experience through web accessibility standards like WCAG 2.1 Level AA (acting as an act of measure for determining if compliance if met). When an organization has to follow ADA Title II or Section 504/508 for digital content, meeting WCAG Level AA serves as the proven way to satisfy the law.

Example: An example of how this works can be taken from a public university that receives federal funding (which makes it subject to section 504) and also receives state funding (making it subject to ADA Title II too) When the university uploads online course materials or updates in the student's portal, those digital systems must be accessible. The law does not specific exactly what technical standards are needed (like contrast, or keyboard navigation); however, the university would use WCAG 2.1 Level AA guidelines (alt text, captions to videos, transcripts, color contrast) to ensure they legally meet their obligations.

Misunderstanding: A common misunderstanding to clarify is that WCAG Level AA is not THE LAW; it is a standard of measure used to help meet the requirements of the law. The law (section 504/508 and ADA Title II) is the obligation, while WCAG 2.1 Level AA standards are the measure.

Sources:

- <https://www.ada.gov/resources/2024-03-08-web-rule/>
- <https://www.section508.gov/>
- <https://www.hhs.gov/civil-rights/for-individuals/disability/index.html>
- <https://www.w3.org/TR/WCAG21/>

Module 02: Color and Contrast

Explain color dependence, text contrast, non-text contrast, and how those claims are tested.

Color Dependence: Color is not used as the only visual means of conveying information, indicating an action, prompting a response, or distinguishing a visual element.

Why it exists: addresses color perception specifically. The goal is that when color is used to convey meaning (selected vs non-selected, showing confirmation, etc.,) that meaning must also be conveyed in an alternative visual way that doesn't depend on the user's ability to perceive color, such as an icon, underline, border, or visible text.

Example: A budget dashboard shows over-budget line items in red text and under-budget items in green text, with no other indicator. A colorblind user can't tell which is which.
(Fix: add an icon next to over-budget rows)

Misunderstanding: A legend elsewhere on the page does not satisfy 1.4.1. The cue has to travel with the element itself. A user shouldn't have to cross-reference a key to understand a single row or field.

Text Contrast: The Visual presentation of text images of text and images of text has a contrast ratio of at least 4.5:1, except for large-scale text and images of large-scale text, which need at least 3:1.

Why it exists: If text and its background are too close in color, some people cannot read it, including people with low vision, color blindness, or age-related vision changes. It is not about looking nice but about whether the text can actually be read.

Example: A hero banner has light grey text on a white background. Run through a contrast checker, that combination is 2.8:1. It may look fine to someone with typical vision.

Misunderstanding: Contrast is measured numerically, not by eyeballing it. Designers with typical vision routinely misjudge what is readable for someone with moderately low vision to color deficiency. A ratio has to actually be calculated.

Non-Text Contrast: The visual presentation of user interface components and graphical objects must have a contrast ratio of at least 3:1 against adjacent colors.

Why it exists: Buttons, icons, and chart elements often rely on faint outlines or color alone instead of text. If those don't stand out enough, some users can't tell a button is there, or can't tell one part of a chart from another. Since there's often no text backup, low contrast here can mean the feature or information is just missed entirely.

Example: A custom checkbox uses a thin light-blue outline (white page) to show it is selected, with no fill change and no checkmark. Measured against the white background, that outline falls under 3:1, so a low-vision user may not be able to tell whether the box is checked at all.

Misunderstanding: The threshold is lower than text contrast. It only applies to visual information that's actually required to identify or operate the component or understand the graphic. A decorative border or a graphic where "a particular presentation is essential to the information isn't covered the same way plain text is.

How they are tested: text and non-text contrast can be both run through WebAIM's contrast check, and it gives you a ratio. Color dependency can be checked using tools like WebAIM's WAVE extension to help flag places color might be carrying meaning, but a person still has to look and judge whether a non-color cue is actually there too.

Sources:

- <https://www.w3.org/WAI/WCAG22/Understanding/use-of-color.html>
- <https://www.w3.org/WAI/WCAG22/Understanding/contrast-minimum.html>
- <https://wave.webaim.org/extension/>
- <https://webaim.org/resources/contrastchecker/>
- <https://www.w3.org/WAI/WCAG22/Understanding/non-text-contrast.html>

Module 03: Complex Visual Information

Explain how charts, maps, diagrams, and infographics can require more than a short text alternative.

Charts, maps, diagrams, and infographics require more than just a short text alternative because they pack a large amount of dense, structured, and multidimensional data into a visual format that cannot fit into a brief 120- character alt-text tag. There is simply too much data to be explained. Instead, complex images may need a longer description that explains the data, relationships, patterns, or other important information shown in the visual.

A long description could be provided in a separate section of the page or through a link. This allowed users to access the detailed information without putting it all into alt text.

Example: A bar chart comparing the number of students enrolled in four courses may contain several categories, values, and trends. The alt text could briefly identify the chart, while a longer description provides the exact values and explains the important relationships shown in the chart.

Common misunderstandings:

Some people believe that by writing a long paragraph inside the alt= "" Attribute, the problem is solved. This is a big no. The reality is that many screen readers don't have a pause button for alt text, and by writing a 500-word paragraph inside of the alt tag, the user had to listen to the entire thing straight without stopping. This means that if they miss a detail or number, they'd have to listen to the entire thing all over again.

Sources:

- <https://www.w3.org/WAI/tutorials/images/complex/>
- <https://educationexpress.uts.edu.au/collections/accessible-practice-for-maths-and-diagrams/resources/add-image-descriptions-for-complex-images/>
- <https://equidox.co/blog/beyond-basic-alt-text-charts-maps-and-diagrams/>